$$\vec{F}_{12} = -G \frac{m_1 m_2}{r_{12}^2} \hat{r}_{12}$$

$$F_G = \frac{G M_T}{(R_T + h')} m$$

$M_T = 6 \times 10^{24}\ Kg$

$R_T = 6.4 \cdot 10^6\ m = 64 \cdot 10^3\ Km$

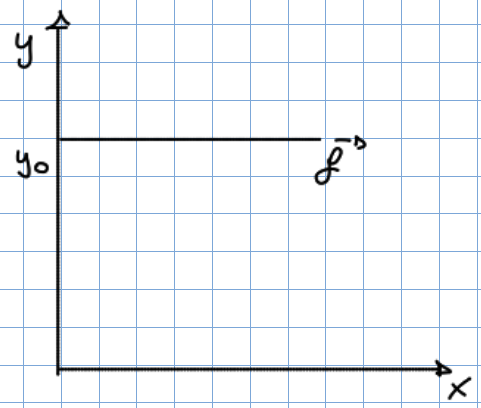
$G \approx 6.67 \cdot 10^{-11}\ \frac{N m^2}{Kg^2}$

$R_T >> h' \Rightarrow R_T + h' \approx R_T$

$$F_G = \left(\frac{G M_T}{R_T^2}\right) m$$

↳ indipendente dal corpo

$$\frac{G M_T}{R_T^2} = \frac{N m^2}{Kg^2} \cdot Kg \times \frac{1}{m^2} = \frac{N}{Kg} = \frac{Kg\, m}{s^2} \cdot \frac{1}{Kg} = \frac{m}{s^2}$$



$$\vec{F} = m_i \vec{a} \quad \vec{F} = \vec{P} = m_g \vec{g} \Rightarrow m_g \vec{g} = m_i \vec{a} \Rightarrow \vec{a} = \frac{m_g}{m_i} \vec{g}$$

$$\Downarrow$$

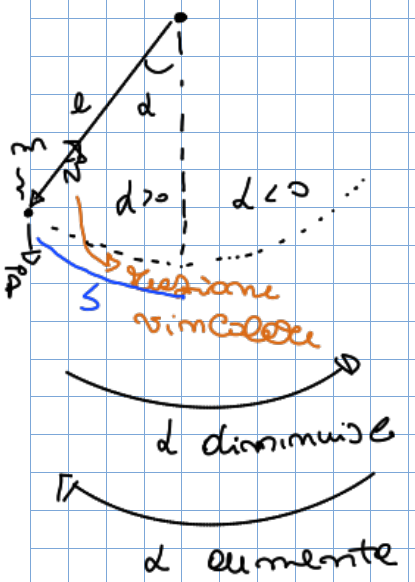
$$y(t) = y_0 - \frac{1}{2} \frac{m}{m_i} g t^2$$

$$\frac{m_g}{m_i} = K$$ ↳ universale

$$\Downarrow$$

$K = 1 \quad \frac{m_g}{m_i} = 1 \Leftrightarrow m_g = m_i$

Pendolo



$$\vec{F} = m\vec{a}$$

$$N - mg\cos\alpha = ma_c = m\frac{v^2}{l}$$

$$a_c = \frac{v^2}{R} = \frac{v^2}{l}$$

$$\begin{cases} \alpha = \alpha(t) \\ v = v(t) \end{cases}$$

$$-mg\,\text{sen}\,\alpha = ma_{\parallel} = \frac{md^2s}{dt^2} \overset{s = l\alpha}{=} ml\frac{d^2\alpha}{dt^2}$$

$$s = l\alpha$$

$$v = \frac{ds}{dt}, \quad a_{\parallel} = \frac{d^2s}{dt^2}$$

$$\Rightarrow l\frac{d^2\alpha}{dt^2} + g\sin\alpha = 0 \Rightarrow \boxed{\frac{d^2\alpha}{dt^2} + \frac{g}{l}\sin\alpha = 0}$$

Equazione moto pendolo semplice

$$\Downarrow$$

$$\alpha = \alpha(t)$$

$$\alpha \to 0 \Rightarrow \sin\alpha \approx \alpha$$

$$\frac{d^2\alpha}{dt^2} + \frac{g}{l}\alpha = 0$$

$$\boxed{\omega^2 = \frac{g}{l}} \Rightarrow \frac{d^2\alpha}{dt^2} + \omega^2\alpha = 0 \Rightarrow \alpha(t) = A\cos(\omega t + \varphi)$$

$$\Downarrow$$

$$\boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}}}$$

$$\alpha_0 = \alpha(t=0)$$

$$v = \frac{ds}{dt} = l\frac{d\alpha}{dt}$$

$$\alpha(t) = \theta \cos(\omega t + \varphi)$$

$$v = \frac{ds}{dt} = \ell \frac{d\alpha}{dt} = -\ell A \omega \sin(\omega t + \varphi)$$

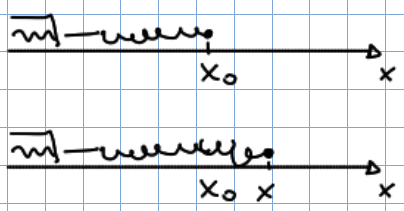
$$a = \frac{d^2 s}{dt^2} = -\ell A \omega^2 \cos(\omega t + \varphi)$$

$$\left.\begin{array}{l} \alpha(t=0) = \alpha_0 \\ v(t_0) = 0 \end{array}\right\} \Rightarrow \alpha(t) = \alpha_0 \cos \omega t$$

$$N - mg\cos\alpha = \frac{mv^2}{\ell}$$

$$N = mg\cos\alpha + m\frac{v^2}{\ell} = mg\cos[\alpha_0 \cos(\omega t)] + m\frac{1}{\ell}\ell^2 \alpha_0^2 \sin^2(\omega t)$$

Forze elastiche



$$\vec{F} = -k(x - x_0)\hat{x}$$

$$\vec{F} = m\vec{a} \Rightarrow F = ma \Rightarrow -k(x - x_0) = m\frac{d^2x}{dt^2}$$

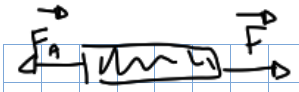
$$\Rightarrow m\frac{d^2x}{dt^2} + k(x - x_0) = 0 \Rightarrow \boxed{\frac{d^2x}{dt^2} + \frac{k}{m}(x - x_0) = 0} \Rightarrow \frac{d^2u}{dt^2} + \frac{k}{m}u = 0$$

$$u = x - x_0 \Rightarrow \frac{du}{dt} = \frac{dx}{dt} \Rightarrow \frac{d^2u}{dt^2} = \frac{d^2x}{dt^2}$$

$$\omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}}$$

$$u(t) = A\cos(\omega t + \varphi) \Rightarrow \boxed{x(t) = x_0 + A\cos(\omega t + \ell)}$$

$$u = x - x_0 \Rightarrow x = x_0 + u$$



$\vec{N} + m\vec{g} = 0 \quad \Rightarrow \quad \vec{N} = -m\vec{g} \qquad N = mg$

$F = \mu_s N$

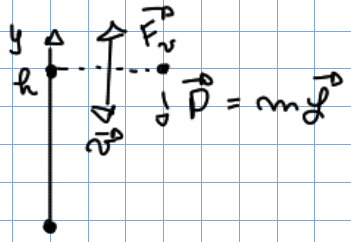
$F > \mu_s N$

$\vec{F} = -\mu_c N \hat{v}$

μ_c coefficiente di attrito cinematico

$\mu_s > \mu_c$

attrito viscoso



$\vec{F}_v = -\beta \vec{v}$

$v(t=0) = 0$

$\vec{F}_v + \vec{P} = m\vec{a}$

$-\beta v_y - mg = m \frac{dv_y}{dt}$ → quanto sulle forze di attrito viscose

$\vec{F}_v + \vec{P} = 0 \quad \Rightarrow \quad \frac{dv_y}{dt} = 0 \quad \Rightarrow \quad -\beta v_y - mg = 0 \quad \Rightarrow \quad v_y = -\frac{mg}{\beta} \equiv v_{y,L}$

$|v_y| = \frac{mg}{\beta}$

$t \to \quad -mg = \frac{dv_y}{dt} \quad \Rightarrow \quad v_y(t) = -gt$

